

Han (Simon) Yan

Ithaca, NY • hyan86652@gmail.com • (206)5199388 • https://hy86652.github.io/Simon_Yan_Portfolio/

EDUCATION

Cornell University

Master of Professional Studies in Information Science, Concentration: **Data Science**

Cumulative GPA: 3.45/4.0

Coursework: Applied Machine Learning · Simulation Modeling & Analysis · Data Communication · Info Computing

University of Washington

Bachelor of Arts in Geography: **Data Science**

Cumulative GPA: 3.55/4.0, Major GPA: **3.89/4.0** | **Annual Dean's List**, 2023 - 2024, 2024 - 2025

Coursework: Data Science · Machine Learning · GIS Database Programming · Elements of Statistical Methods

WORK EXPERIENCE

Central China Securities Co Ltd (HKEX: 1375)

Shanghai, China

Quantitative Investment Researcher Intern (Data Engineering & Quantitative Modeling)

06/2024 - 08/2024

- **Optimized 10+** multi-factor alpha signals strategies (e.g., Momentum, Volatility) by incorporating market regime classification to enhance signal stability and reduce drawdown during volatile periods.
- **Engineered** a modular backtesting framework in **Python**, enabling the automated simulation of **20+** equity strategies to support the team's hypothesis validation and parameter tuning.
- **Built** an automated ETL pipeline (Python/Pandas) processing **100K+** daily records, implementing data validation and outlier-handling logic to ensure consistent, analysis-ready data for downstream reporting.
- **Translated** complex performance metrics into clear, actionable briefs for non-technical stakeholders, directly informing committee-level resource allocation decisions

Central China Securities Co Ltd (HKEX: 1375)

Hongkong, China

Financial Product Researcher Intern (Market & Business Analysis)

06/2023 - 08/2023

- **Spearheaded** comprehensive sector research across Semiconductor and Autonomous Driving industries, producing 3 deep-dive investment memos that directly informed the team's product development roadmap.
- **Assessed** historical performance and risk-adjusted return profiles of 4 candidate funds against benchmark indices, informing selection of 2 funds for the firm's FOF allocation

Projects

Private Equity Forecast Analysis | The Riverside Company, Quantitative Analyst Lead

09/2025 - 12/2025

- **Led** team on a four-part investment performance study for a private equity client, evaluating projection accuracy, holding-period dynamics, and early-warning signals across the firm's portfolio.
- **Modeled** investment outcomes via logistic regression, identifying a **~54%** early-miss threshold beyond which negative-IRR probability surpasses **50%**, and an optimal **2-4** year asset holding period.
- **Realigned** project strategy based on client feedback, translating complex statistical methodologies into outcome-first reporting to drive non-technical stakeholder decision-making.
- **Delivered** an MVP dashboard and technical documentation to ensure reproducible underwriting workflows.

Wikipedia RAG - ChatBot | Cornell Course Project

10/2025 - 12/2025

- **Led a 3-member team** to architect and deploy an interactive RAG web chatting application (R, Shiny), delivering an end-to-end verifiable knowledge retrieval workflow **to mitigate LLM hallucinations**.
- **Engineered** an automated query pipeline integrating **OpenAI and MediaWiki APIs**, utilizing a source-grounded prompt design to systematically **reduce LLM hallucinations**.
- **Developed** a scalable, reactive UI featuring session history and exportable citations, prioritizing transparent source tracking and ease of use for non-technical users.

Predicting National Flood Insurance Program Claims | Cornell Course Project

08/2025 - 10/2025

- **Built and benchmarked** predictive regression models (XGBoost, Random Forest, Lasso) in R on a **500K+** FEMA claims dataset to forecast flood payouts, achieving an **R² of 0.956**, reducing baseline RMSE by **80%**.
- Used **SHAP** and **Partial Dependence Plots** to identify and communicate key risk drivers in plain-language terms, providing actionable insights for insurance pricing and underwriting decisions.

ADDITIONAL INFORMATION

Languages: Python, R, Java, SQL, JavaScript, HTML/CSS, Shell Script

Libraries & Tools: Pandas, NumPy, Scikit-learn, Shiny, Matplotlib, Seaborn, Git, Tableau, Docker, A/B Test

Methods & Tools: Excel (Pivot Tables, VLOOKUP), PowerPoint, Regression Modeling, Data Visualization